# Do Now · retrieval

## Half adders and full adders

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

When we group four neighbouring 1s on a Karnaugh map, what happens to the Boolean expression?

### Q2

Represent −13 using 8-bit sign and magnitude.

### Q3

Insert 40, 20, 60, 10, 30, 50 into an empty binary search tree. Draw the resulting tree.

### Q4

Using a 5-bit two’s complement mantissa and 4-bit two’s complement exponent, calculate A − B. Show exponent alignment and mantissa subtraction. A = $0.1100_2 \times 2^3$; B = $0.1000_2 \times 2^2$.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Start with binary addition — One binary column can produce a sum bit and a carry bit.

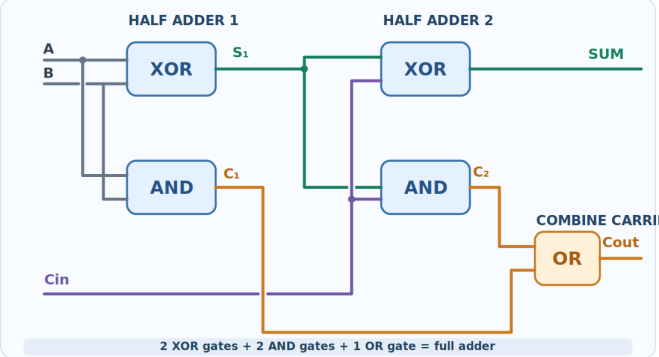
## Core idea 2

Discover the two gates — XOR calculates Sum; AND calculates Carry.

## Core idea 3

Find the limitation — What happens when a previous column sends a carry into this column?

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Core idea 4

First half adder — Add A and B to produce an intermediate Sum $S_1$ and Carry $C_1$.

## Core idea 5

Combine both carries — Feed $C_1$ and $C_2$ into an OR gate to obtain Carry-out.

## Vocabulary · use the correct technical terms

half adder — A circuit adding two input bits.

full adder — A circuit adding two bits plus a carry-in.

Sum — The result bit for the current column.

Carry — A bit passed to the next column.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Model 1 + 1 with a half adder

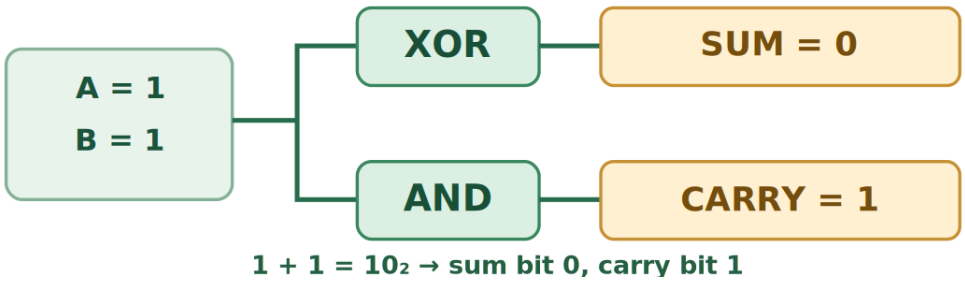
## Follow the worked method

1. Rightmost result bit — CODE:
Sum = 0

2. Trace XOR and AND — CODE:
1 XOR 1 = 0 → Sum
1 AND 1 = 1 → Carry

3. Connect both half adders — CODE:
$S_1$ = A XOR B
Sum = $S_1$ XOR Cin
Cout = (A AND B) OR ($S_1$ AND Cin)

**Source model · final illustrated step**




**1 + 1 = $10_2$ → sum bit 0, carry bit 1**

## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Complete the same formulae
CODE:
S = A ? B
C = A ? B

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

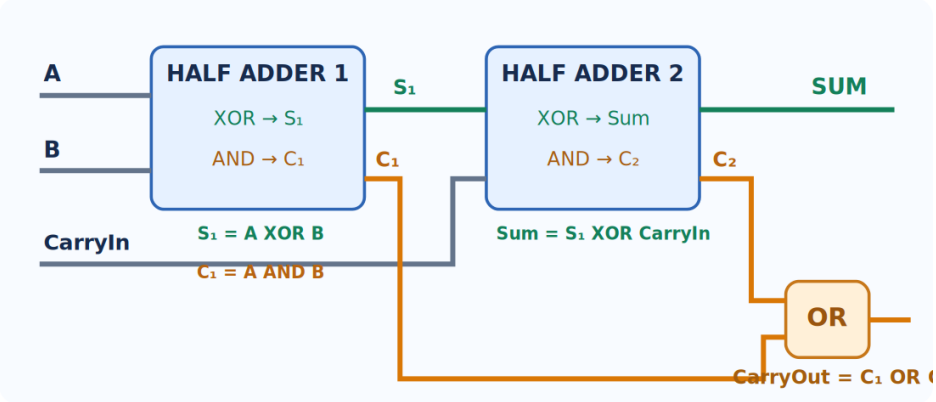
Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Half-adder question

State the Sum/Carry expressions, complete the truth table, then explain how half adders contribute to a full adder.

**Question 1 · original source diagram**




Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 2

Add 0111 + 0001

Use bit-by-bit addition and identify where the carry travels.

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

For a half adder, Sum is produced by XOR and Carry is produced by AND.

## Use key vocabulary from memory

1. Explain half adder in your own words.

2. Explain full adder in your own words.

3. Explain Sum in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.